

EE 423 Capstone Presentation

By Team 6

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Senior Level Electrical Engineering

Project Goals:



- Intended Design: Studio monitors
- Inspiration: Yamaha HS7 studio monitors
- Approximately 1 cubic foot in volume
- Fairly high quality sound for affordable to mid-price range
- Desired 40 Hz-20 kHz frequency response, while staying well under our budget of \$375

Project Focus:

- Vented box design chosen to keep volume size to desired volume of 1 cubic foot
- Crossover network to keep response as flat as possible with a crossover frequency around 1.6 kHz

Project Results:

- Vented box, 16" height, 9.5" width, 11" length
- Total External Volume: .97 cubic ft, 27.4 liters
- Total Internal Volume: .69 cubic ft, 19.4 liters
- Budget: \$137.13 per box (\$138.52 under budget!)

Drivers Considered

- Drivers chosen based on recommended design volume.
- Two Goldwood woofers considered, (6.5" & 8" diameters)
Ultimately, we decided on a 6.5" Pyramid woofer as it provided the best quality compared to its cost (\$19.50)
- Various Dayton tweeters considered. Ultimately, a 1 1/8" silk dome tweeter was chosen, again because of the value of its performance compared to cost (\$19.75)

Pyramid WX65X 6-½" White Poly Woofer



6.5" Pyramid woofer provided great quality compared to its relatively low cost of \$19.50

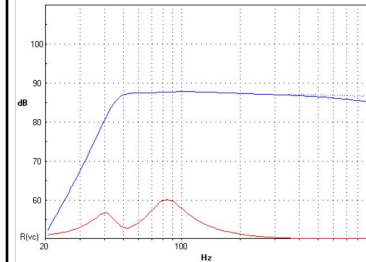
Woofer Specifications:

Product Specifications

Nominal Diameter	6-1/2"
Power Handling (RMS)	150 Watts
Power Handling (max)	300 Watts
Impedance	4 ohms
Frequency Response	70 to 3,500 Hz
Sensitivity	87.1 dB 1W/1m
Voice Coil Diameter	1"
Magnet Weight	40 oz.

Thiele-Small Parameters

Resonant Frequency (Fs)	69.5 Hz
DC Resistance (Re)	3.3 ohms
Mechanical Q (Qms)	2.03
Electromagnetic Q (Qes)	0.82
Total Q (Qts)	0.58
Compliance Equivalent Volume (Vas)	0.3 ft. ³
Maximum Linear Excursion (Xmax)	2 mm



Pyramid surprisingly offers no response curves for their products. These curves were obtained using Boxwrx. We felt confident enough to proceed with this woofer.

Dayton Audio DC28F-8 1-⅛" Silk Dome Tweeter



Highly rated tweeter of a small diameter

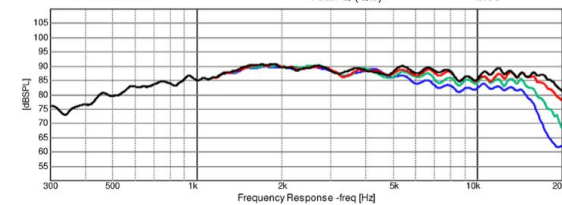
Tweeter Specifications:

Product Specifications

Tweeter Type	Soft Dome
Cone / Dome Diameter	1-1/8"
Impedance	8 ohms
Power Handling (RMS)	50 Watts
Frequency Response	1,300 to 20,000 Hz
Sensitivity	89 dB 1W/1m

Thiele-Small Parameters

Resonant Frequency (Fs)	834 Hz
DC Resistance (Re)	5.40 ohms
Voice Coil Inductance (Le)	0.09 mH
Mechanical Q (Qms)	0.81
Electromagnetic Q (Qes)	1.33
Total Q (Qts)	0.50



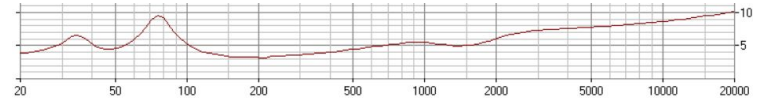
Tweeter was rated very highly and, upon inspection, seemed like it would crossover well with the woofer. This actually ended up being more difficult than anticipated.

Construction



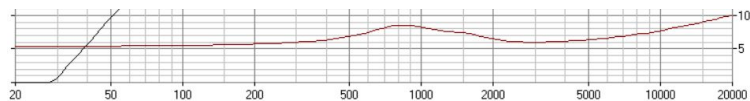
Port tube was 2.5" in diameter, which required 7.1" length to tune to 50 Hz.

Testing - Woofer



Resonance frequency f_s [Hz]	62,3458	Linear excursion per side [mm]	2
Electrical Q-factor Q_{es}	1,054	DC resistance R_{dc} [ohm]	3,3
Mechanical Q-factor Q_{ms}	3,5213	Inductance of voice coil L_e [mH]	0,2565
equivalent volume V_{as} [liter]	8,495	substitution resistor R_{e2} [ohm]	0,001
Effective piston area [cm ²]	210	substitution inductance L_{e2} [mH]	0,001
P max [W]	300	substitution resistor R_{e3} [ohm]	0,001
Nominal impedance [ohm]	4	substitution inductance L_{e3} [mH]	0,001

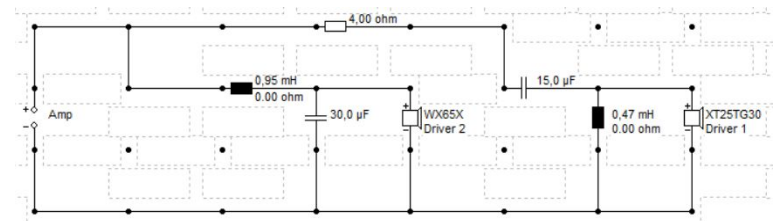
Testing - Tweeter



Tweeter Measured Parameters:

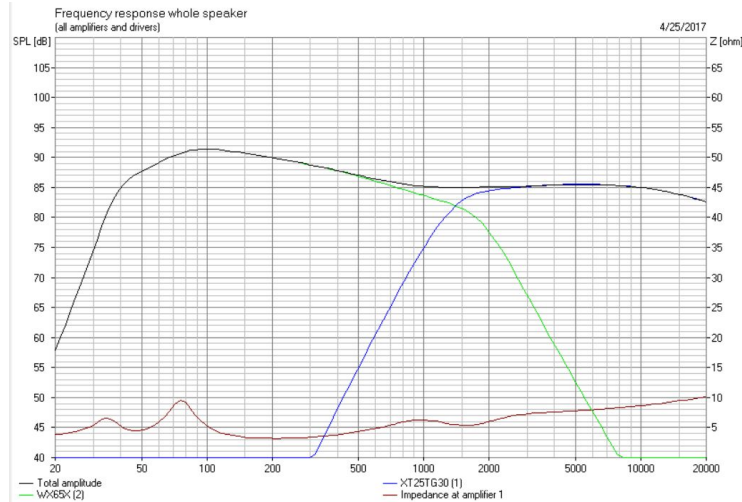
Resonance frequency	834	Linear excursion	0.5
Electrical Q factor	1.33	DC resistance	5.4
Mechanical Q factor	0.81	Voice coil inductance	0.07
Equivalent volume	0.01	Substitution resistor	0.001
Piston area	6.6	Substitution inductor	0.001
Max. power	50	Substitution resistor	0.001
Nom. impedance	8	Substitution inductor	0.001

Crossover Network Design

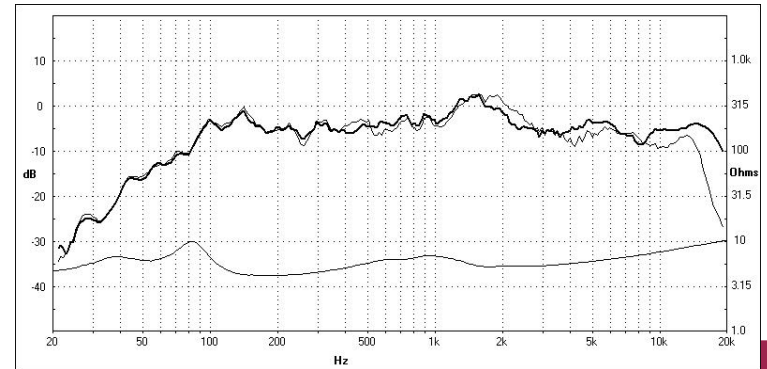


Crossover frequency tuned to 1600 Hz with the help of the Boxtsim program. The woofer and tweeter were having phase issues regardless of polarity, so some significant tweaking was needed. With professor's help, we achieved a relatively flat response.

Simulated System Performance



System Performance



System Cost

Items	Part #	Quantity	Price	Total
Dayton Audio DC28F-8 1-1/8" Silk Dome Tweeter	275-070	2	\$19.75	\$39.50
Pyramid WX65X 6-1/2" White Poly Woofer	290-270	2	\$17.93	\$35.86
Parts Express #6 x 1" Deep Thread Pan Head Screw	081-440	1	\$3.10	\$3.10
Parts Express Speaker Cabinet Port Tube 2-1/2" ID	260-386	2	\$3.18	\$6.36
Acousta-Stuf Polyfill Speaker Cabinet Sound Dampin	260-317	1	\$9.00	\$9.00
Square Speaker Wire Terminal Cup 4" Gold Spring-L	260-279	2	\$5.50	\$11.00
Dayton Audio PMPC-15 15uF 250V Precision Audio	027-248	2	\$10.68	\$21.36
Jantzen Audio 0.95mH 18 AWG Air Core Inductor Cr	255-248	2	\$9.92	\$19.84
Jantzen Audio 0.47mH 15 AWG Air Core Inductor Cr	255-408	2	\$11.52	\$23.04
Mills 4 Ohm 12W Non-Inductive Resistor	005-4	2	\$4.25	\$8.50
Dayton Audio PMPC-30 30uF 250V Precision Audio	027-258	2	\$18.72	\$37.44
Black Perforated Large Hole Crossover Board Pair 5	260-194	1	\$3.50	\$3.50
Estimated Lumber (per square foot)	N/A	10.79	\$3.50	\$37.77
True Value Varnish - Red Mahogany	N/A	2	\$8.99	\$17.98
			Total	\$274.25
			Allotted	\$375.00
			Balance	\$138.52



Future Work:

- With more money, we would have purchased a higher quality woofer, since we ended up missing more frequencies than our simulation led us to believe.
- With more time, we would have built a subwoofer, and made amplifiers built in the speakers.
- For an actual production prototype, we would source our crossover components from another vendor, as these components consumed half of our budget.

